Web Technology I

Unit 1 - Internet and Web

Unit 1 - Internet and Web [5 Hrs.]

1.1. History of Internet and Web

1.2. Uses of Internet and Services

1.3. Introduction to WWW

1.4. Component of WWW (Web, Webpage, Website, Homepage, Web Browsers, Web Servers, URL and Search Engines)

1.5. Types of Web Pages & its Processing in WWW

1.6. Internet protocols (TCP/IP, ARP, HTTP, FTP, SMTP, POP, SNMP) and applications

Web Technology

https://www.geeksforgeeks.org/web-technology/

https://www.halvorsen.blog/documents/programming/web/web.php

Learning Web Technology is essential today because Internet has become the number one source to information, and many of the traditional software applications have become Web Applications. Web Applications have become more powerful and can fully replace desktop application in most situations.

That's why you need to know basic Web Programming, including HTML, CSS and JavaScript. To create more powerful Web Sites and Web Applications you also need to know about Web Servers, Database Systems and Web Frameworks like PHP, ASP.NET, etc.

It all started with Internet (1960s) and the World Wide Web - WWW (1991). The first Web Browser, Netscape, came in 1994. This was the beginning of a new era, where everything is connected on internet, the so-called Internet of Things (IoT).

Web Technology refers to the various tools and techniques that are utilized in the process of communication between different types of devices over the internet. A web browser is used to access web pages.

(Web technologies refers to the way computers/devices communicate with each other using markup languages. It is communication across the web, and create, deliver or manage web content using hypertext markup language (HTML). A web page is a web document which is written in HTML.)

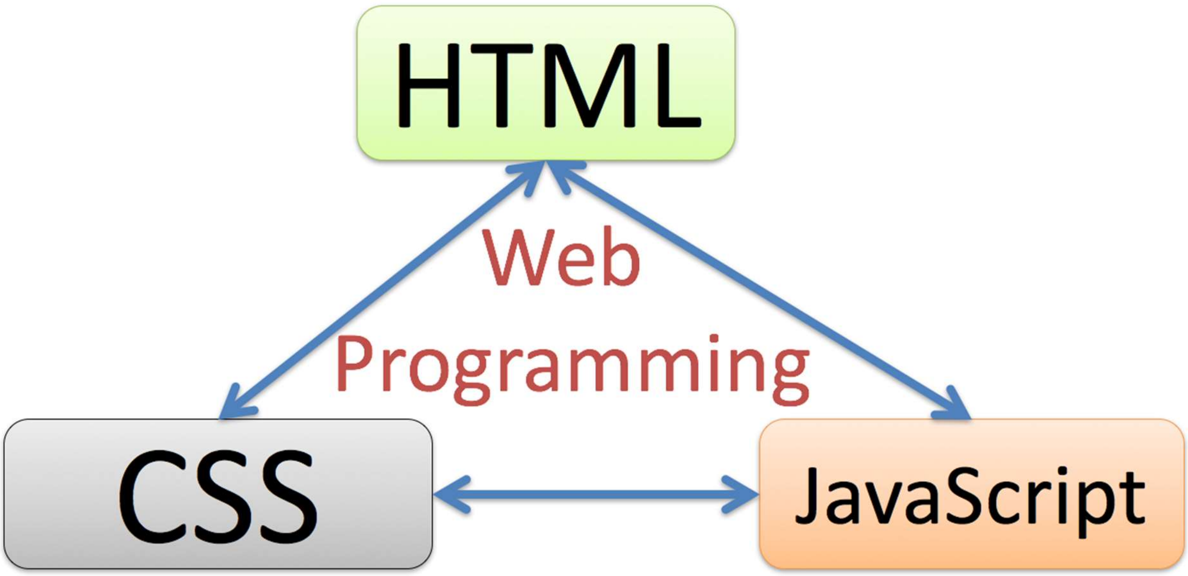
To create a typical web page, you need to combine 3 languages:

HTML - The language for building web pages

CSS - The language for styling web pages

JavaScript - The language for programming web pages

Below we see the triangle used in most web pages, namely HTML, CSS and JavaScript.



We use **HTML** to define the content of web pages, **CSS** is used to specify the layout of web pages, while **JavaScript** is used to program the behaviour of web pages.

For creating more dynamic web pages, we typically also use a web framework like PHP or ASP.NET, etc. With these frameworks you can communicate with a database for storing or retrieving data.

Internet and Web

https://www.geeksforgeeks.org/the-internet-and-the-web/

The terms Internet and the Web (World Wide Web) are synonymous in the minds of many, but they have different meanings.

Internet –

The Internet: In simplest terms, the Internet is a global network comprised of smaller networks that are interconnected using **standardized** communication protocols.

The **Internet** is a massive network of networks that connects millions of computers worldwide.

Internet is a global communication system that links together thousands of individual networks. It allows exchange of information between two or more computers on a network. Thus, internet helps in transfer of messages through mail, chat, video & audio conference, etc. It has become mandatory for day-to-day activities: bills payment, online shopping and surfing, tutoring, working, communicating with peers, etc.

Computers connected to the Internet can communicate with one another with a number of protocols such as HTTP (Hypertext Transfer Protocol), SMTP (Simple Mail Transfer Protocol), FTP (File Transfer Protocol), IRC (Internet relay chat), IM (instant messaging), Telnet, and P2P (peer-to-peer).

Extra –

The Internet: In simplest terms, the Internet is a global network comprised of smaller networks that are interconnected using **standardized** communication protocols. The Internet standards describe a framework known as the Internet protocol suite. This model divides methods into a *layered system of protocols.*
These layers are as follows:

1. **Application layer (highest) –** concerned with the data(URL, type, etc.). This is where HTTP, HTTPS, etc., comes in.
2. **Transport layer –** responsible for end-to-end communication over a network.
3. **Network layer –** provides data route.

The Internet provides a variety of information and communication facilities; contains forums, databases, email, hypertext, etc. It consists of private, public, academic, business, and government networks of local to global scope, linked by a broad array of electronic, wireless, and optical networking technologies.

Web -

The **World Wide Web** is a system of interlinked hypertext documents and programs that can be accessed via the Internet primarily by using HTTP.

The World Wide Web: The Web is the only way to access information through the Internet. It's a system of Internet servers that support specially formatted documents. The documents are formatted in a markup language called **HTML**, or "HyperText Markup Language", which supports a number of features including links and multimedia. These documents are interlinked using hypertext links and are accessible via the Internet.

To link hypertext to the Internet, we need:

1. The markup language, i.e., HTML.
2. The transfer protocol, e.g., HTTP.
3. Uniform Resource Locator (URL), the address of the resource.

We access the Web using **Web browsers**.

Assignment -

What's difference between The Internet and The Web ?

https://www.geeksforgeeks.org/whats-difference-internet-web/

The Internet is a global network of networks while the Web, also referred to formally as World Wide Web (www) is a collection of information that is accessed via the Internet. Another way to look at this difference is, the Internet is infrastructure while the Web is served on top of that infrastructure. Alternatively, the Internet can be viewed as a big book store while the Web can be viewed as a collection of books on that store. At a high level, we can even think of the Internet as hardware and the Web as software!

Difference between Web and Internet:

Internet	Web
The Internet is the network of networks and the network allows to exchange of data between two or more computers.	The Web is a way to access information through the Internet.
It is also known as the Network of Networks.	The Web is a model for sharing information using the Internet.
The Internet is a way of transporting information between devices.	The protocol used by the web is HTTP.
Accessible in a variety of ways.	The Web is accessed by the Web Browser.
Network protocols are used to transport data.	Accesses documents and online sites through browsers.
Global network of networks	Collection of interconnected websites

Internet	Web
Access Can be accessed using various devices	Accessed through a web browser
Connectivity Network of networks that allows devices to communicate and exchange data	Connectivity Allows users to access and view web pages, multimedia content, and other resources over the Internet
Protocols TCP/IP, FTP, SMTP, POP3, etc.	Protocols HTTP, HTTPS, FTP, SMTP, etc.
Infrastructure Consists of routers, switches, servers, and other networking hardware	Infrastructure Consists of web servers, web browsers, and other software and hardware
Used for communication, sharing of resources, and accessing information from around the world	Used for publishing and accessing web pages, multimedia content, and other resources on the Internet
No single creator	Creator Tim Berners-Lee
Provides the underlying infrastructure for the Web, email, and other online services	Provides a platform for publishing and accessing information and resources on the Internet

Assignment –

Advantages and disadvantages of Internet and web.

1.1. History of Internet and Web

https://news.geeksforgeeks.org/knowledge/history-of-internet

https://www.encyclopedia.com/economics/encyclopedias-almanacs-transcripts-and-maps/history-internet-and-world-wide-web-www

The **Internet** is a global network of interconnected computer networks that communicate with each other over the Internet Protocol Suite (TCP/IP). It is a network of networks made up of private, public, academic, and government networks ranging from local to global in extent and connected by a diverse set of electronic, wireless, and optical networking technologies. The Internet holds a huge range of information as well as services like Email, Voice-over IP, Television, Games, File Sharing, Shopping, etc.

The World Wide Web (WWW), commonly known as the **Web**, is an information system where documents and other web resources are identified by Uniform Resource Locators (URLs, such as https://www.example.com/), which may be interlinked by hypertext, and are accessible over the Internet. The resources of the WWW may be accessed by users by a software application called a web browser.

History of Internet and web –

https://www.computerhope.com/history/internet.htm

Father of Web - Computer Scientist **Tim Berners-Lee**

The first workable prototype of the Internet came in the late 1960s with the creation of ARPANET, or the Advanced Research Projects Agency Network. Originally funded by the U.S. Department of Defense, ARPANET used packet switching to allow multiple computers to communicate on a single network. The technology continued to grow in the 1970s after scientists Robert Kahn and Vinton Cerf developed Transmission Control Protocol and Internet Protocol, or TCP/IP, a communications model that set standards for how data could be transmitted between multiple networks. ARPANET adopted TCP/IP on January 1, 1983, and from there researchers began to assemble the “network of networks” that became the modern Internet. The online world then took on a more recognizable form in 1990, when computer scientist Tim Berners-Lee invented the World Wide Web. While it’s often confused with the Internet itself, the web is actually just the most common means of accessing data online in the form of websites and hyperlinks. The web helped popularize the Internet among the public, and served as a crucial step in developing the vast trove of information that most of us now access on a daily basis.

English scientist Tim Berners-Lee invented the World Wide Web in 1989. He wrote the first web browser in 1990 while employed at The European Organization for Nuclear Research (CERN - Conseil Europeen pour la Recherche Nuclaire) near Geneva, Switzerland. The browser was released outside CERN in 1991, first to other research institutions starting in January 1991 and then to the general public in August 1991. The World Wide Web has been central to the development of the Information Age and is the primary tool billions of people use to interact on the Internet.

History of the Web

- Internet (1960s)
- World Wide Web - WWW (1991)
- First Web Browser - Netscape, 1994
- Google, 1998
- Facebook, 2004
- Smartphones, 2007
- Tablets, 2010

Extra –

https://www.computerhope.com/jargon/c/cern.htm

https://home.cern/

1.2. Uses of Internet and Services

https://www.brainkart.com/article/Services-Available-on-the-Internet_36837/

https://www.geeksforgeeks.org/internet-and-its-services/

https://quicklearncomputer.com/services-of-internet/

Based on a recent survey of Internet traffic, the 10 most popular uses of the Internet are -

1. **Electronic mail**
   At least 85% of the inhabitants of cyberspace send and receive e-mail. Some 20 million e-mail messages cross the Internet every week.
2. **Research**
3. **Downloading files**
4. **Discussion groups**
5. **Interactive games**
6. **Education and self-improvement**
   On-line courses and workshops have found yet another outlet.
7. **Friendship and dating**
   You may be surprised at the number of electronic "personals" that you can find on the World Wide Web.
8. **Electronic newspapers and magazines**
   This category includes late-breaking news, weather, and sports. We're likely to see this category leap to the top five in the next several years.
9. **Job-hunting**
   Classified ads are in abundance, but most are for technical positions.

10. Shopping

It's difficult to believe that this category even ranks. It appears that "cybermalls" are more for curious than serious shoppers.

1.3. Introduction to WWW

https://www.geeksforgeeks.org/world-wide-web-www/

https://www.brainkart.com/article/The-Role-of-WWW-as-a-Service-on-the-Internet_36838/

The **World Wide Web** is abbreviated as WWW and is commonly known as the web. The WWW was initiated by CERN (European library for Nuclear Research) in 1989. WWW can be defined as the collection of different websites around the world, containing different information shared via local servers(or computers).

History:

It is a project created, by Timothy Berner Lee in 1989, for researchers to work together effectively at CERN. is an organization, named the World Wide Web Consortium (W3C), which was developed for further development of the web. This organization is directed by Tim Berner's Lee, aka the father of the web.

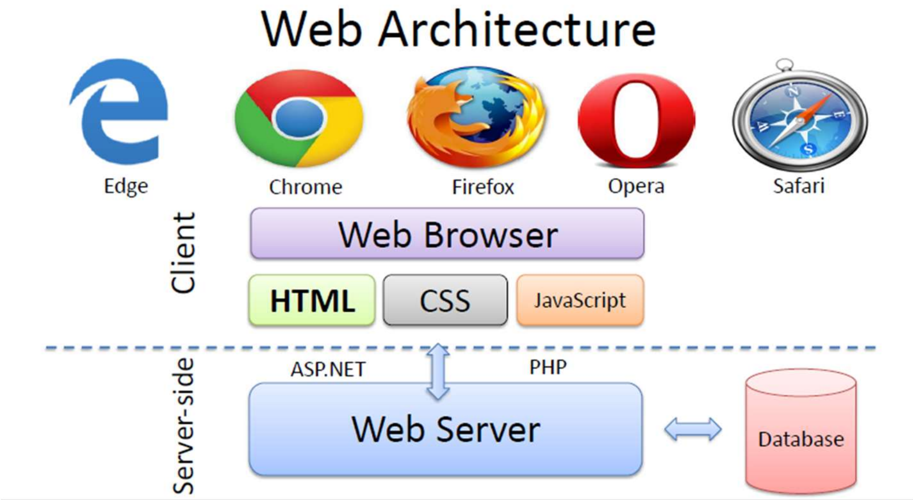
Working of WWW:

The World Wide Web is based on several different technologies: Web browsers, Hypertext Markup Language (HTML) and Hypertext Transfer Protocol (HTTP).

A Web browser is used to access web pages. Web browsers can be defined as programs which display text, data, pictures, animation and video on the Internet. Hyperlinked resources on the World Wide Web can be accessed using software interfaces provided by Web browsers. Initially, Web browsers were used only for surfing the Web but now they have become more universal. Web browsers can be used for several tasks including conducting searches, mailing, transferring files, and much more. Some of the commonly used browsers are Internet Explorer, Opera Mini, and Google Chrome.

Features of WWW:

- HyperText Information System
- Cross-Platform
- Distributed
- Open Standards and Open Source
- Uses Web Browsers to provide a single interface for many services
- Dynamic, Interactive and Evolving.



1.4. Component of WWW (Web, Webpage, Website, Homepage, Web Browsers, Web Servers, URL and Search Engines)

Components of the Web: There are 3 components of the web:

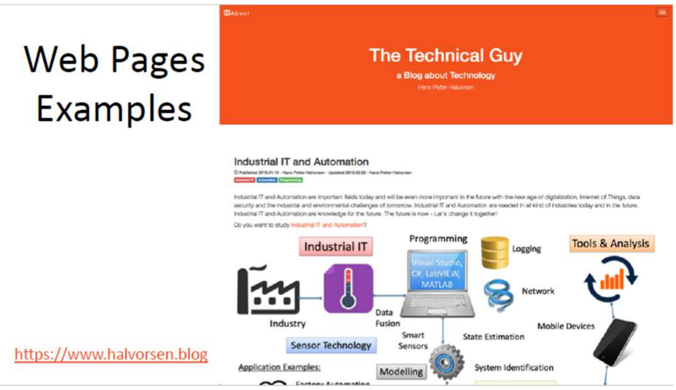
1. **Uniform Resource Locator (URL):** serves as a system for resources on the web.
2. **HyperText Transfer Protocol (HTTP):** specifies communication of browser and server.
3. **Hyper Text Markup Language (HTML):** defines the structure, organisation and content of a webpage.

Webpage –

https://www.computerhope.com/jargon/w/webpage.htm

Webpage is a document commonly written in Hypertext Markup Language (HTML) that is accessible (displayed) through the Internet using an Internet browser (Web browser such as Firefox, Google Chrome, Opera, Microsoft Internet Explorer or Edge, or Apple's Safari) . A web page is accessed by entering a URL address and may contain text, graphics and hyperlinks to other web pages and files. . These are also often called just "pages.

The Web Pages are stored on the Web Server and sent to the client on request from the Web Browser



Website –

https://www.geeksforgeeks.org/what-is-a-website/

https://www.computerhope.com/jargon/w/website.htm

https://www.geeksforgeeks.org/different-types-of-websites/

A website is a collection of web pages (web pages are digital files that are written using HTML(HyperText Markup Language)) which are grouped together and usually connected together in various ways. Often called a "web site" or simply a "site.“

There are different types of websites on the whole internet, we had chosen some most common categories to give you a brief idea –

- **Blogs:** These types of websites are managed by an individual or a small group of persons, they can cover any topics — they can give you fashion tips, music tips, travel tips, fitness tips. Nowadays professional blogging has become an external popular way of earning money online.
- **E-commerce:** These websites are well known as online shops. These websites allow us to make purchasing products and online payments for products and services. Stores can be handled as standalone websites.
- **Portfolio:** These types of websites acts as an extension of a freelancer resume. It provides a convenient way for potential clients to view your work while also allowing you to expand on your skills or services.
- **Brochure:** These types of websites are mainly used by small businesses, these types of websites act as a digital business card, and used to display contact information, and to advertise services, with just a few pages.
- **News and Magazines:** These websites needs less explanation, the main purpose of these types of websites is to keep their readers up-to-date from current affairs whereas magazines focus on the entertainment.
- **Social Media:** We all know about some famous social media websites like Facebook, Twitter, Reddit, and many more. These websites are usually created to let people share their thoughts, images, videos, and other useful components.
- **Educational:** Educational websites are quite simple to understand as their name itself explains it. These websites are designed to display information via audio or videos or images.

- **Portal:** These types of websites are used for internal purposes within the school, institute, or any business, These websites often contain a login process allowing students to access their credential information or allows employees to access their emails and alerts.

Homepage –

Home page is a very common and important part of a webpage. It is the first webpage that appears when a visitor visits the website. The home page of a website is very important as it sets the look and feel of the website and directs viewers to the rest of the pages on the website.

Web Browsers –

https://www.tutorialspoint.com/computer_concepts/computer_concepts_web_browsing_software.htm

The web browser (also called browser) is an application software to explore www (World Wide Web). It provides an interface between the server and the client and requests to the server for web documents and services.

A web browser is an application program for retrieving, presenting and traversing information resources on the World Wide Web.

An information resource (web data) is identified by a Uniform Resource Identifier (URI/URL) that may be a web page, image, video or other piece of content available in web server.

Some examples of web browser are:

- Internet Explorer
- Google Chrome,
- Firefox,
- Opera,
- Safari

The Web Browser

Table 15.2 Browsers and Vendors

Browser	Vendor
Internet Explorer	Microsoft
Google Chrome	Google
Mozilla Firefox	Mozilla
Netscape Navigator	Netscape Communications Corp.
Opera	Opera Software
Safari	Apple
Sea Monkey	Mozilla Foundation
K-meleon	K-meleon

Web Servers –

Web server is a program (computer system application) which processes the network requests via HTTP of the users and serves them with files (web pages). The term can refer to the entire system, or specifically to the software that accepts and supervises the HTTP requests.

A special high-end computer that hosts a website on the Internet. Today we have Cloud services that act as web servers.

A Web server is a program or computer machine that generates and transmits responses to client requests for Web resources using HTTP protocol.

Every computer on the Internet that contains a Web site must have a Web server program.

Any computer can be turned into a Web server by installing server software and connecting the machine to the Internet.

Table 15.3 Web Server

S.N.	Web Server
1	Apache IITTP Server.
2.	Internet Information Services (IIS)
3.	Sun Java System Web Server

Some Examples of Web Server

Apache HTTP Server:

- The Apache HTTP Server, commonly referred to as Apache is a web server software program which gave the initial boost for the growth of the World Wide Web.
- In 2009, it became the first web server software to surpass the 100 million website milestone.

- The server is aimed at serving a great deal of widely popular modern web platforms/operating systems such as Unix, Windows, Linux, Solaris, Novell NetWare, FreeBSD, Mac OS X, Microsoft Windows, OS/2, etc.

Microsoft Internet Information Services (IIS):

- IIS is one of the components of Microsoft Windows and is Microsoft's implementation of a web server.
- The protocols supported include HTTP, HTTPS, FTP, FTPS & SMTP .
- It's estimated that around 25% of all websites utilize IIS.

URL –

- URL stands for Uniform Resource Locator.
- A URL is a formatted text string used by Web browsers, email clients and other software to identify a network resource on the Internet.
- Network resources are files that can be plain Web pages, other text documents, graphics, or programs.

URL strings consist of three parts (*substrings*):

1. network protocol: Typical URL protocols include
http:// and ftp:// .
2. host name or address: identifies a computer or other network device. Hosts come from standard Internet databases such as DNS and can be names or IP addresses.
3. file or resource location:
These substrings are separated by special characters as follows:
protocol :// host / location

Example:

http://www.uneca.org/acs

Search Engines –

https://www.tutorialspoint.com/computer_concepts/computer_concepts_search_engines.htm

Search Engine is an application (software system) that allows you to search (or is designed to search) for content/Information on the web (www). It displays multiple web pages based on the content or a word you have typed. A huge information available on internet on various topics. The information may be a mix of web pages, images, and other types of files. Some search engines also mine data available in databases or open directories. Search engines also maintain real-time information by running an algorithm on a web crawler.

The most popular search engines are : -

- Google

- Bing
- Yahoo
- Ask
- AOL - America Online

Figure 15.10 Search Engines

Extra –

Domain Name:

A domain name is an identification a string that defines a area of administrative autonomy, authority or control within the Internet. Domain names are formed by the rules and procedures of the Domain Name System (DNS). Any name registered in the DNS is a domain name. Domain names are used in various networking backgrounds and application-specific naming and addressing purposes. In general, a domain name represents an Internet Protocol (IP) resource, such as a personal computer used to access the Internet, a server computer hosting a web site.

There are several domain names available:

• Generic domain names such as **.com, .edu, .gov, .net.**

• Country level domain names such as **au, in, za, us.**

The following table shows the Generic Top-Level Domain names:

Table 15.1 Top-Level Domain names

Domain Name	Meaning
Com	Commercial business
edu	Education
gov	U.S. Government agency
int	International entity
mil	U.S. military

Web Hosting:

Hosting is the location where the website is physically located. Group of webpages (linked webpages) licensed to be called a website only when the webpage is hosted on the webserver. The webserver is a set of files transmitted to user computers when they specify the website's address.

Web Facilitating is an administration of give online space to capacity of site pages .These Site pages are made accessible by means of WWW.The organizations which offer site facilitating are known as web host.

Examples of Web Hosting Companies:

• Go Daddy

• Amazon Web service

• Digital Ocean

• Free webhostingarea.com

Assignment –

Difference between website and webpage

1.5. Types of Web Pages & its Processing in WWW

https://www.tutorialspoint.com/internet_technologies/web_pages.htm

web page is a document available on world wide web. Web Pages are stored on web server and can be viewed using a web browser.

A web page can contain huge information including text, graphics, audio, video and hyper links. These hyper links are the link to other web pages.

Collection of linked web pages on a web server is known as **website**. There is unique Uniform Resource Locator (URL) is associated with each web page.

Types of Web Pages –

1. **Static Web Pages**
2. **Dynamic Web Pages**

Static Web Pages -

Static pages show the same content each time they are viewed. Static Web pages are returned by the server which are prebuilt source code files built using simple languages such as HTML, CSS, or JavaScript. There is no processing of content on the server (according to the user) in Static Webpages/Websites. Web pages are returned by the server with no change therefore, static Websites are fast. There is no interaction with databases. Also, they are less costly as the host does not need to support server-side processing with different languages.

Dynamic pages have content that can change each time they are accessed. Dynamic pages are typically written in scripting languages such as PHP, Perl, ASP, or JSP. The scripts in the pages run functions on the server that return things like the date and time, and database information.

Dynamic Web pages are returned by the server which is processed during runtime means they are not prebuilt web pages, but they are built during runtime according to the user's demand with the help of server-side scripting languages such as PHP, Node.js, ASP.NET and many more supported by the server. So, they are slower than static websites but updates and interaction with databases are possible. Dynamic Webpages/Websites are used over Static Websites as updates can be done very easily as compared to static websites (Where altering in every page is required) but in Dynamic Websites, it is possible to do a common change once, and it will reflect in all the web pages.

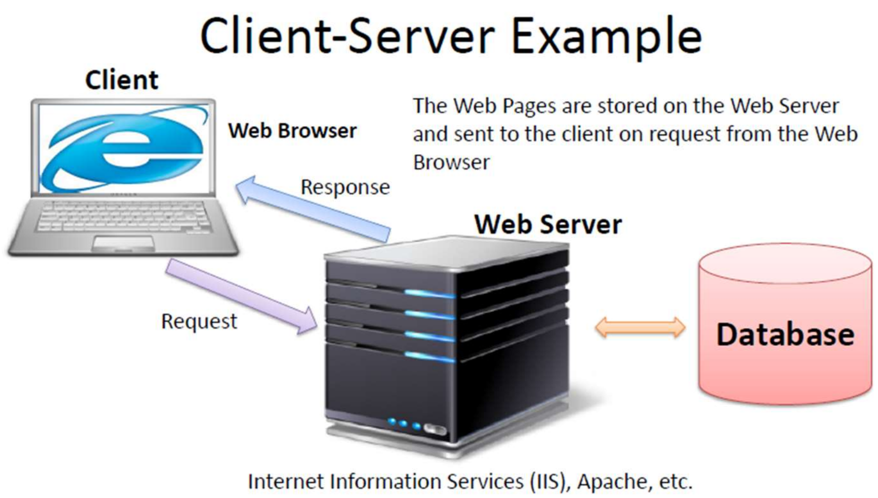
Other types of web pages –

https://www.unleashed-technologies.com/blog/8-types-webpages-every-website-should-have-2021

- Homepage
- Product Page
- About Page
- Contact Page
- Search Page
- Custom 404 Page

Note -

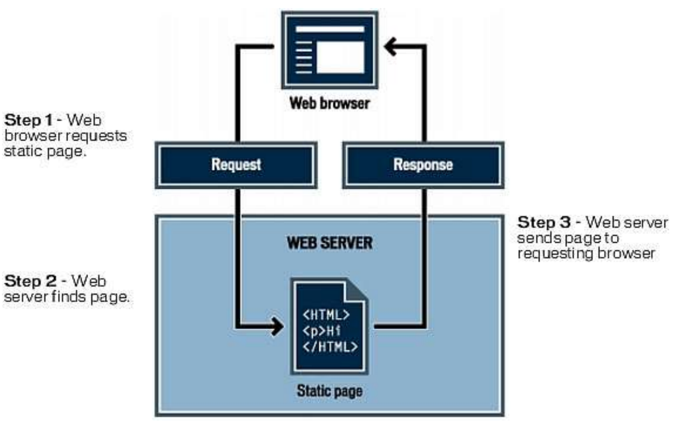
All the information is returned as HTML code, so when the page gets to your browser, all the browser has to do is translate the HTML.



Processing static web pages

A web server is software that serves web pages in response to requests from web browsers. A page request is generated when a visitor clicks a link on a web page, selects a bookmark in a browser, or enters a URL in a browser's address text box.

When the web server receives a request for a static page, the server reads the request, finds the page, and sends it to the requesting browser, as shown in the following figure:



In the case of web applications, certain lines of code are undetermined when the visitor requests the page. These lines must be determined by some mechanism before the page can be sent to the browser.

Extra -

Fig - Process of retrieving a static webpage.

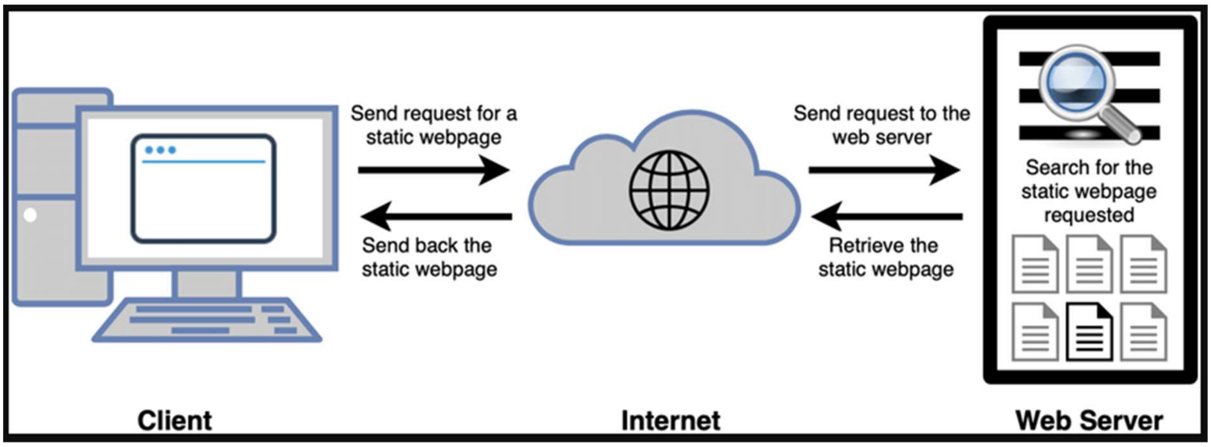
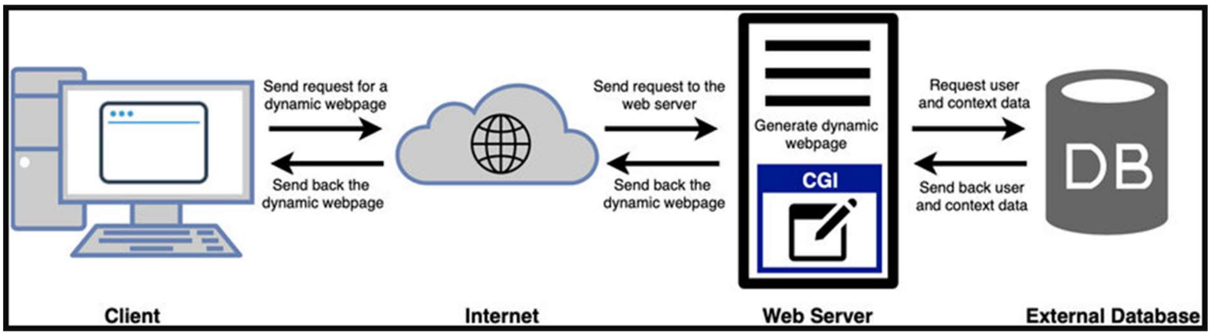


Fig - Process of retrieving a dynamic webpage.



Assignment –

Difference between Static and dynamic Webpage/websites.

Advantages and disadvantages of static and dynamic Webpage/websites

https://teleporthq.io/blog/static-vs-dynamic-websites-learn-the-differences

1.6. Internet protocols (TCP/IP, ARP, HTTP, FTP, SMTP, POP, SNMP) and applications

https://tyonote.com/internet_protocols_types/

https://www.geeksforgeeks.org/types-of-internet-protocols/

https://www.ccexpert.us/operating-systems/describe-internet-protocols-and-applications.html

A **protocol** is a set of rules. **Internet Protocols** are a set of rules that governs the communication and exchange of data over the internet. Both the sender and receiver should follow the same protocols in order to communicate the data.

In order to understand it better, let's take an example of a language. Any language has its own set of vocabulary and grammar which we need to know if we want to communicate in that language. Similarly, over the internet whenever we access a website or exchange some data with another device then these processes are governed by a set of rules called the internet protocols.

Working of Internet Protocol

The internet and many other data networks work by organizing data into small pieces called packets. Each large data sent between two network devices is divided into smaller packets by the underlying hardware and software. Each network protocol defines the rules for how its data packets must be organized in specific ways according to the protocols the network supports.

Timing is crucial to network operation. Protocols require messages to arrive within a certain amount of time so that computers do not wait indefinitely for messages that may have been lost. Therefore, systems maintain one or more timers during transmission of data. Protocols also initiate alternative actions if the network does not meet the timing rules. Many protocols consist of a suite of other protocols that are stacked in layers. These layers depend on the operation of the other layers in the suite to function properly.

Why are protocols needed in computer networks?

Protocols are essential for computer networks because they provide standards for communication between different devices. Protocols ensure that the data sent from one device can be properly decoded and understood by the receiving device. They also provide accurate and efficient data transmission between different devices and help establish reliable connections.

Need of Protocols

It may be that the sender and receiver of data are parts of different networks, located in different parts of the world having different data transfer rates. So, we need protocols to manage the flow control of data, and access control of the link being shared in the communication channel. Suppose there is a sender X who has a data transmission rate of 10 Mbps. And, there is a receiver Y who has a data receiving rate of 5Mbps. Since the rate of receiving the data is slow so some data will be lost during transmission. In order to avoid this, receiver Y needs to inform sender X about the speed mismatch so that sender X can adjust its transmission rate. Similarly, the access control decides the node which will access the link shared in the communication channel at a particular instant in time. If not the transmitted data will collide if many computers send data simultaneously through the same link resulting in the corruption or loss of data.

Functions of protocols:

- Identifying errors
- Compressing the data
- Deciding how the data is to be sent.
- Addressing the data
- Deciding how to announce sent and received data.

TCP/IP

(Transmission Control Protocol/Internet Protocol) - Transports data on the Internet

These are a set of standard rules that allows different types of computers to communicate with each other.

The IP ensures that each computer that is connected to the Internet is having a specific serial number called the IP address.

TCP specifies how data is exchanged over the internet and how it should be broken into IP packets. It also makes sure that the packets have information about the source of the message data, the destination of the message data, the sequence in which the message data should be re-assembled and checks if the message has been sent correctly to the specific destination. The TCP is also known as a connection-oriented protocol.

Note –

Transmission control protocol is used for communication over a network. In TCP data is broken down into small packets and then sent to the destination. However, IP is making sure packets are transmitted to the right address.

ARP

(Address Resolution Protocol)

In order to map an IP address into a hardware address, the computer uses the ARP protocol which broadcasts a request message that contains an IP address to which the target computer replies with both the original IP address and the hardware address.

HTTP

(Hypertext transfer protocol) - Defines how files are exchanged on the web

A protocol used to transfer hypertext pages across World Wide Web (WWW).

This protocol is used to transfer hypertexts over the internet and it is defined by the www(world wide web) for information transfer. This protocol defines how the information needs to be formatted and transmitted. And, it also defines the various actions the web browsers should take in response to the calls made to access a particular web page. Whenever a user opens their web browser, the user will indirectly use HTTP as this is the protocol that is being used to share text, images, and other multimedia files on the World Wide Web.

Note –

HTTP is based on client and server model. HTTP is used for making a connection between the web client and web server. HTTP shows information in web pages.

HTTPS (HyperText Transfer Protocol Secure) –

HTTPS is an extension of the Hypertext Transfer Protocol (HTTP). It is used for secure communication over a computer network with the SSL/TLS protocol for encryption and authentication. So, generally, a website has an HTTP protocol but if the website is such that it receives some sensitive information such as credit card details, debit card details, OTP, etc then it requires an SSL certificate installed to make the website more secure. So, before entering any sensitive information on a website, we should check if the link is HTTPS or not. If it is not HTTPS then it may not be secure enough to enter sensitive information.

FTP

(File Transfer Protocol) - Provides services for file transfer and manipulation

File transfer protocol enables transferring of text and binary files over transmission control protocol (TCP) connection. FTP allows transferring files according to a strict mechanism of ownership and access restrictions.

This protocol is used for transferring files from one system to the other. This works on a client-server model. When a machine requests for file transfer from another machine, the FTO sets up a connection between the two and authenticates each other using their ID and Password. And, the desired file transfer takes place between the machines.

SMTP

(Simple mail transfer protocol) - Sends mail in a TCP/IP network

This protocol is dedicated to sending e-mail messages originated on a local host over a TCP connection to a remote server. SMTP defines the set of rules which shows two programs to send and receive mail over the network.

These protocols are important for sending and distributing outgoing emails. This protocol uses the header of the mail to get the email id of the receiver and enters the mail into the queue of outgoing mail. And as soon as it delivers the mail to the receiving email id, it removes the email from the outgoing list. The message or the electronic mail may consider the text, video, image, etc. It helps in setting up some communication server rules.

POP

(Post Office Protocol) - POP3 - Downloads e-mail messages from an e-mail server

POP3 stands for Post Office Protocol version 3. It has two Message Access Agents (MAAs) where one is client MAA (Message Access Agent) and another is server MAA(Message Access Agent) for accessing the messages from the mailbox. This protocol helps us to retrieve and manage emails from the mailbox on the receiver mail server to the receiver's computer. This is implied between the receiver and the receiver mail server. It can also be called a one-way client-server protocol. The POP3 WORKS ON THE 2 PORTS I.E. PORT 110 AND PORT 995.

SNMP

(Simple network management protocol)

A simple protocol that defines messages related to network management. Through the use of a simple network management protocol (SNMP) any host on the LAN can configure network devices such as routers.

Extra –

https://www.khanacademy.org/computing/computers-and-internet/xcae6f4a7ff015e7d:the-internet/xcae6f4a7ff015e7d:the-internet-protocol-suite/a/the-internet-protocols